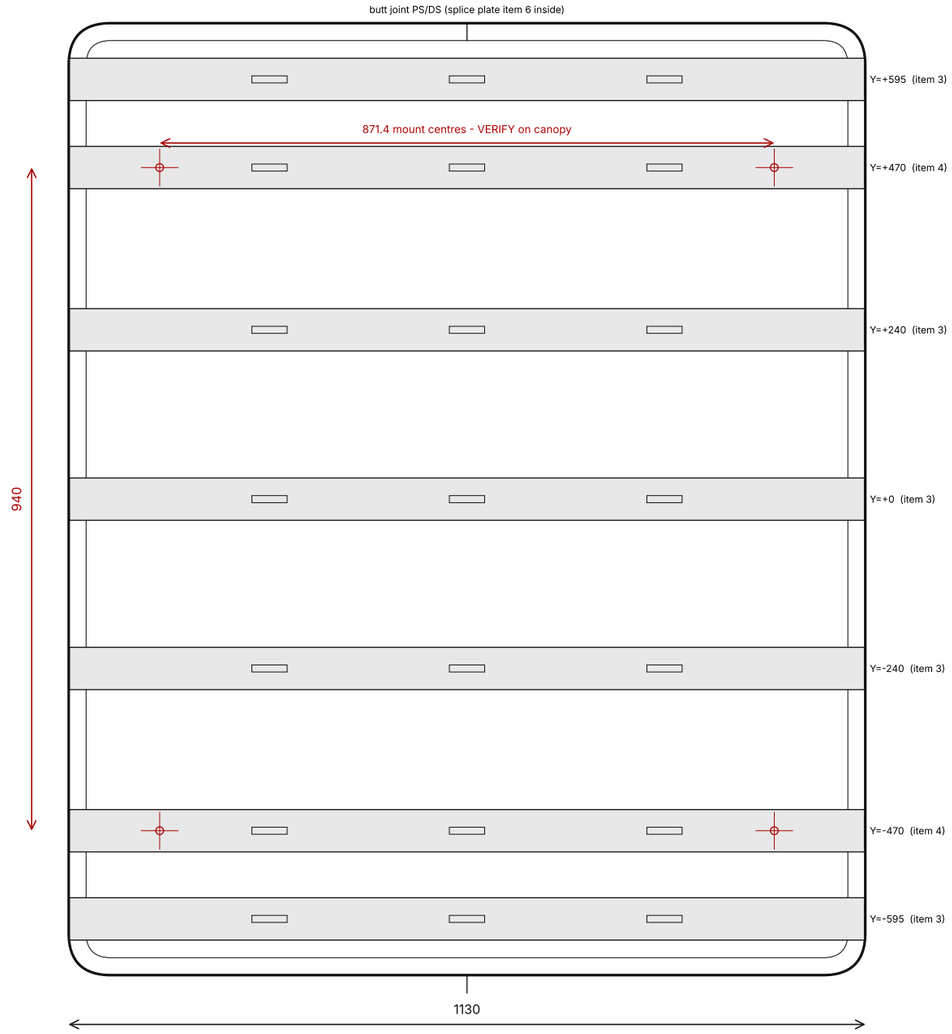
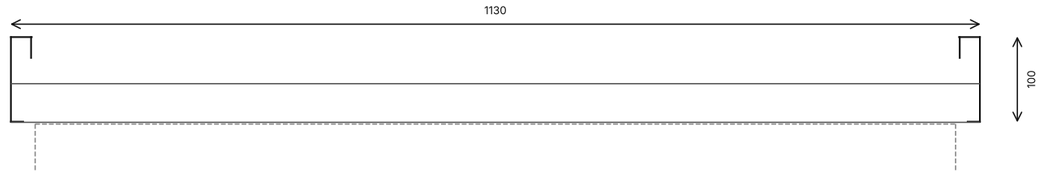
PLAN VIEW (from above) scale 1:8.7



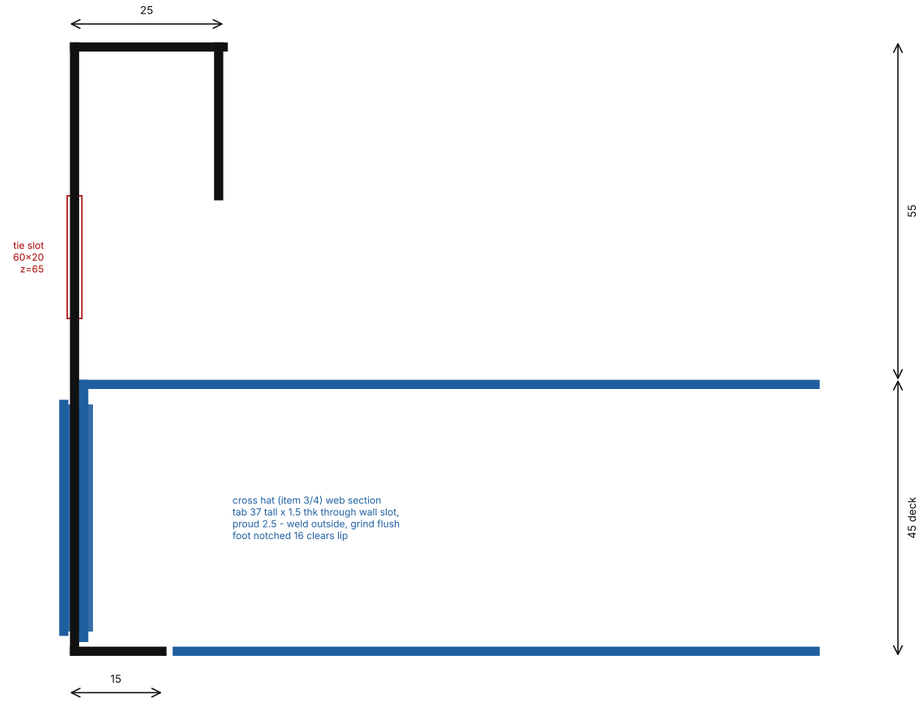
front of vehicle = +Y (top of view). Rack symmetric front/rear. Item 1 = left (PS) half, item 2 = right (DS) half.

SECTION A-A (looking along vehicle) scale 1:7.1



canopy roof 40105 (1072 wide, side flanges down) - rack walls overhang roof edge by 29 each side

DETAIL B - wall / cross member end (scale 1:1)



KEY DATA

• Overall 1130 (W) x 1350 (L) x 100 (H), plan corners R60 - same footprint as Rev-02, bolts to existing canopy roof holes.

• Deck at Z=45 (7 top-hat crosses 60×45), wall lip 55 above deck, boxed top rail (25 flange + 25 return).

• Material: 304 (1.4301) 2B, 1.5 mm throughout. Fabricated mass ≈ 26.6 kg. Rated payload 250 kg evenly distributed (see RRD-2000 calc report).

• Rated tie-down points: deck accessory slots (3 per cross, 21 total), 150 kg each. Wall slots: lashing / cargo net only (≤ 25 kg lateral, 150 kg vertical).

• Mount: 4x M10 at X=±435.7, Y=±470 through mount crosses (item 4) into canopy roof rib holes. VERIFY hole pitch on canopy before cutting item 4.

COMPANY VELD ARMOUR - Hilux GD6 DC canopy roof rack		DWG NO. **RRD-2000-Rev-03**
TITLE **Roof Rack DC - cage style - GA**	MATERIAL 304 (1.4301) 2B all parts	THICKNESS 1.5 mm
QTY / ASSY 1	FINISH Brush / passivate after weld	MASS (each) 26.65 kg
SCALE 1:8.7	DATE / REV 2026-10-20 Rev-03	SHEET 1 of 9 A3

Unless otherwise stated: dimensions in mm. Linear ±0.5, bend angle ±1°, flat pattern per K=0.44, inside R1.5. Deburr and break sharp edges. Do not scale drawing.

BILL OF MATERIALS - RRD-2000-Rev-03

ITEM	PART NO.	DESCRIPTION	QTY	MASS kg	DXF / SOURCE
1	RRD-2001-Rev-03	Side rail PS (left half): long side + 2 corners + 2 half end walls	1	4.54	RRD-2001 FLAT.dxf
2	RRD-2002-Rev-03	Side rail DS (right half) - mirror of item 1	1	4.54	RRD-2002 FLAT.dxf
3	RRD-2003-Rev-03	Deck cross member, top-hat 60×45, at Y = 0, ±240, ±595	5	2.28	RRD-2003 FLAT.dxf
4	RRD-2004-Rev-03	Mount cross member, as item 3 + mount slots, at Y = ±470	2	2.28	RRD-2004 FLAT.dxf
5	RRD-2005-Rev-03	Mount cross closing plate 54 × 1000 (makes item 4 a box section)	2	0.64	RRD-2005 FLAT.dxf
6	RRD-2006-Rev-03	Rail splice plate 60 × 90	2	0.07	RRD-2006 FLAT.dxf
7	RRD-2007-Rev-03	Roof backing plate 120 × 30, slot 24 × 11	4	0.04	RRD-2007 FLAT.dxf
8	bought	M10 × 110 A2-70 hex bolt + nyloc nut + 2x Ø30 × 3 washers	4		hardware
9	bought	Spacer tube Ø16 × 1.5 × 20.5 long, 304 (between closing plate and roof rib)	4		hardware
10	bought	Crush tube Ø12.7 × 1.6 × 42 long, 304 (welded inside item 4)	4		hardware
11	bought	TIG filler ER308L; pickling paste / passivation	-		consumable

Total fabricated mass ≈ 26.6 kg (excl. hardware). One 2500 × 1250 × 1.5 sheet nests the complete set (see RRD-2000 NEST dxf, ~71% utilisation).

ASSEMBLY SEQUENCE

1. Laser cut all parts from the DXF files (layer CUT). Deburr. Press-brake per each part sheet (all bends DOWN as drawn, inside R1.5). Check first-off of items 1 and 3 against the formed section before running the batch.
2. Item 4: slide closing plate (5) inside the hat flush with the feet, stitch weld 25/100 both webs from below. Weld crush tubes (10) between closing plate hole and top slot.
3. Rails (1)(2): roll both corner zones in-plane around a R60 former (web only - lip, flange and return are V-notched). Close every V-notch seam with TIG, grind flush outside.
4. Jig halves on a flat table, outside faces to stops: tack PS to DS at front and rear centre butt joints. Fit splice plates (6) inside the wall across each joint; full weld joints inside and outside, grind outside flush.
5. Insert the 7 crosses from inside: tabs through the wall slots, deck faces flush at Z=45, feet on the table. Measure diagonals (equal within 1.5 mm) and width 1130 at every cross before welding.
6. Weld every tab on the outside of the wall (3 mm fillet full tab height, both tabs each end), grind flush. Stitch weld 20/150 cross feet-to-lip where they touch (optional) and the mount crosses full length.
7. Clean, pickle and passivate (or brush and passivate). Fit rubber/EPDM strip 3 mm under all cross feet and the lip to protect the canopy paint.
8. Fit to canopy: M10 bolt (8) down through mount cross slot, crush tube, spacer (9), roof rib hole, backing plate (7) inside the canopy, washer, nyloc. Torque 35 Nm. Re-torque after 500 km.

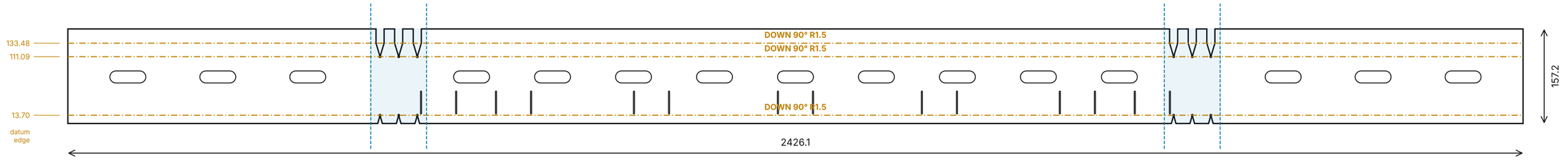
LOAD RATING

- Rated payload 250 kg static, evenly distributed on the deck; dynamic design factor 3.0 g vertical (off-road), 1.5 g braking, 1.0 g lateral - see calculation report RRD-2000-CALC.
- Tie down to the deck accessory slots (rated 150 kg per slot). Wall slots are for lashing and cargo nets only - do not ratchet heavy loads against the wall (plate yield limit ≈ 25 kg lateral per slot).
- Check the canopy roof panel (40105) and canopy side walls for the 250 kg + 27 kg rack load - this report covers the rack only.

Unless otherwise stated: dimensions in mm. Linear ±0.5, bend angle ±1°, flat pattern per K=0.44, inside R1.5. Deburr and break sharp edges. Do not scale drawing.

COMPANY VELD ARMOUR - Hilux GD6 DC canopy roof rack		DWG NO. **RRD-2000-Rev-03**
TITLE **Roof Rack DC - BOM & assembly notes**	MATERIAL 304 (1.4301) 2B all parts	THICKNESS 1.5 mm
QTY / ASSY 1	FINISH Brush / passivate after weld	MASS (each) 26.65 kg
SCALE -	DATE / REV 2026-10-20 Rev-03	SHEET 2 of 9 A3

FLAT PATTERN (viewed from OUTSIDE / top face - all bends DOWN) scale 1:7



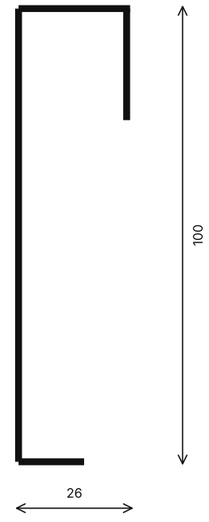
BEND TABLE

No.	Pos. from datum edge	Angle	Dir.	Inside R	Formed (outside)
1	13.70	90°	DOWN	R1.5	Lip 15
2	111.09	90°	DOWN	R1.5	Wall 100
3	133.48	90°	DOWN	R1.5	Top flange 25
	edge 157.18				Return 25

K-factor 0.44, BA(90°)=3.39, BD(90°)=2.61. Adjust flat to your tooling if your BD differs.

Bend sequence: 1 then 2, 3 (long bends first, lip/return last). Check first-off against FORMED SECTION.

FORMED SECTION (scale 0.8:1)



Lip 15 | Wall 100 | Top flange 25 | Return 25 (outside dims)

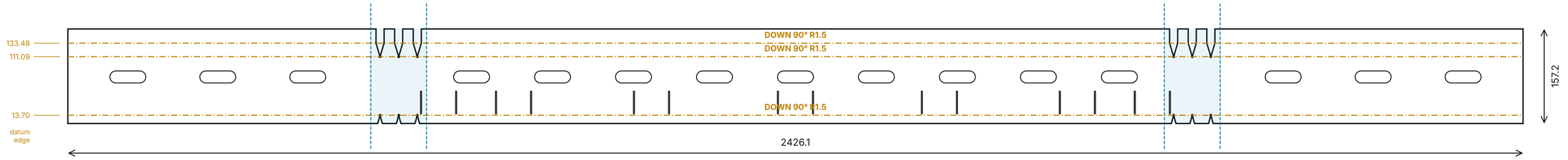
NOTES

1. Developed length 2426.1: 2 corners rolled in-plane to outside R60 AFTER press-brake bends.
2. Corner zones (hatched) : web only is rolled; weld V-notch seams closed in lip, flange and return, grind flush.
3. Two halves (PS + DS) butt-weld at front and rear centre line; fit splice plate RRD-2006 behind joint.
4. Tab slots 1.8 × 38.0 receive cross-member tabs (RRD-2003/2004) - weld tabs outside, grind flush.
5. DS rail is the mirror image: same profile, mirrored flat (bend DOWN viewed from outside face).
6. Material: 304 stainless 2B, 1.5 mm. Laser cut from DXF 'RRD-2001-Rev-03_Side_Rail_PS_left_FLAT.dxf' (1:1, mm), layer CUT only; BEND/ROLL/NOTES layers are reference.

Unless otherwise stated: dimensions in mm. Linear ±0.5, bend angle ±1°, flat pattern per K=0.44, inside R1.5. Deburr and break sharp edges. Do not scale drawing.

COMPANY		DWG NO.
VELD ARMOUR - Hilux GD6 DC canopy roof rack		**RRD-2001-Rev-03**
TITLE	MATERIAL	THICKNESS
Side Rail PS (left)	304 (1.4301) 2B	1.5 mm
QTY / ASSY	FINISH	MASS (each)
1	Brush / passivate after weld	4.50 kg
SCALE	DATE / REV	SHEET
1:7	2026-10-20 Rev-03	3 of 9 A3

FLAT PATTERN (viewed from OUTSIDE / top face - all bends DOWN) scale 1:7



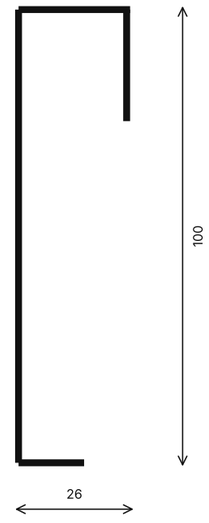
BEND TABLE

No.	Pos. from datum edge	Angle	Dir.	Inside R	Formed (outside)
1	13.70	90°	DOWN	R1.5	Lip 15
2	111.09	90°	DOWN	R1.5	Wall 100
3	133.48	90°	DOWN	R1.5	Top flange 25
	edge 157.18				Return 25

K-factor 0.44, BA(90°)=3.39, BD(90°)=2.61. Adjust flat to your tooling if your BD differs.

Bend sequence: 1 then 2, 3 (long bends first, lip/return last). Check first-off against FORMED SECTION.

FORMED SECTION (scale 0.8:1)



Lip 15 | Wall 100 | Top flange 25 | Return 25 (outside dims)

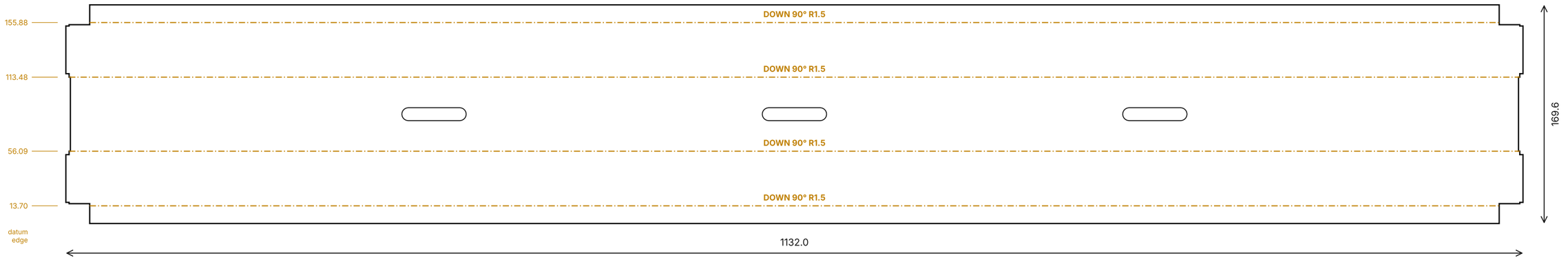
NOTES

1. Developed length 2426.1: 2 corners rolled in-plane to outside R60 AFTER press-brake bends.
2. Corner zones (hatched) : web only is rolled; weld V-notch seams closed in lip, flange and return, grind flush.
3. Two halves (PS + DS) butt-weld at front and rear centre line; fit splice plate RRD-2006 behind joint.
4. Tab slots 1.8 × 38.0 receive cross-member tabs (RRD-2003/2004) - weld tabs outside, grind flush.
5. DS rail is the mirror image: same profile, mirrored flat (bend DOWN viewed from outside face).
6. Material: 304 stainless 2B, 1.5 mm. Laser cut from DXF 'RRD-2002-Rev-03_Side_Rail_DS_right_FLAT.dxf' (1:1, mm), layer CUT only; BEND/ROLL/NOTES layers are reference.

Unless otherwise stated: dimensions in mm. Linear ±0.5, bend angle ±1°, flat pattern per K=0.44, inside R1.5. Deburr and break sharp edges. Do not scale drawing.

COMPANY VELD ARMOUR - Hilux GD6 DC canopy roof rack		DWG NO. **RRD-2002-Rev-03**
TITLE **Side Rail DS (right)**	MATERIAL 304 (1.4301) 2B	THICKNESS 1.5 mm
QTY / ASSY 1	FINISH Brush / passivate after weld	MASS (each) 4.50 kg
SCALE 1:7	DATE / REV 2026-10-20 Rev-03	SHEET 4 of 9 A3

FLAT PATTERN (viewed from OUTSIDE / top face - all bends DOWN) scale 1:3



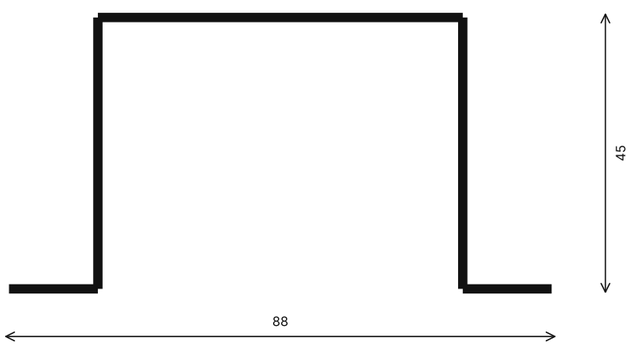
BEND TABLE

No.	Pos. from datum edge	Angle	Dir.	Inside R	Formed (outside)
1	13.70	90°	DOWN	R1.5	Foot 15
2	56.09	90°	DOWN	R1.5	Web 45
3	113.48	90°	DOWN	R1.5	Top 60
4	155.88	90°	DOWN	R1.5	Web 45
	edge 169.57				Foot 15

K-factor 0.44, BA(90°)=3.39, BD(90°)=2.61. Adjust flat to your tooling if your BD differs.

Bend sequence: 1 then 2, 3, 4 (long bends first, lip/return last). Check first-off against FORMED SECTION.

FORMED SECTION (scale 1.0:1)



Foot 15 | Web 45 | Top 60 | Web 45 | Foot 15 (outside dims)

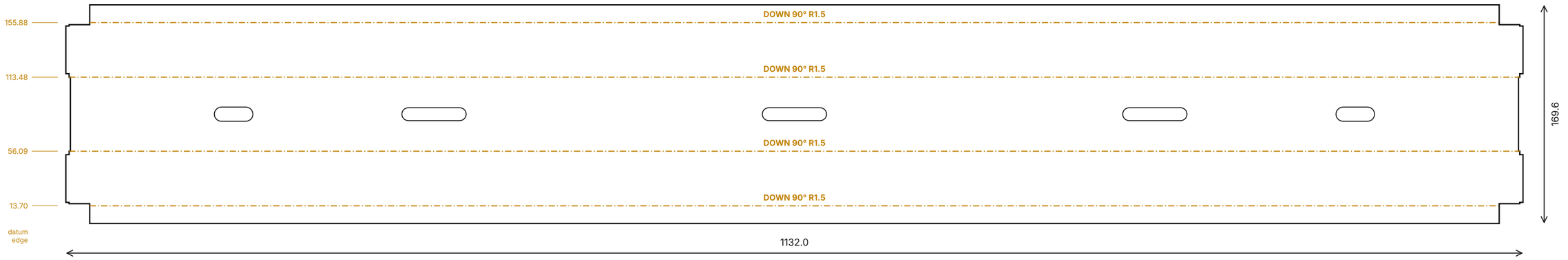
NOTES

1. Top-hat section, open side DOWN (feet rest on canopy roof skin).
2. End tabs (4 off, 37 × 1.5) pass through side-rail wall slots; weld outside and grind flush.
3. Feet notched 16 mm at ends to clear side-rail bottom lip.
4. 3 x accessory slots 50 × 10 in deck face (rated tie-down points).
5. Material: 304 stainless 2B, 1.5 mm. Laser cut from DXF 'RRD-2003-Rev-03_Deck_Cross_Member_FLAT.dxf' (1:1, mm), layer CUT only; BEND/ROLL/NOTES layers are reference.

Unless otherwise stated: dimensions in mm. Linear ±0.5, bend angle ±1°, flat pattern per K=0.44, inside R1.5. Deburr and break sharp edges. Do not scale drawing.

COMPANY VELD ARMOUR - Hilux GD6 DC canopy roof rack		DWG NO. **RRD-2003-Rev-03**
TITLE **Deck Cross Member**	MATERIAL 304 (1.4301) 2B	THICKNESS 1.5 mm
QTY / ASSY 5	FINISH Brush / passivate after weld	MASS (each) 2.26 kg
SCALE 1:3	DATE / REV 2026-10-20 Rev-03	SHEET 5 of 9 A3

FLAT PATTERN (viewed from OUTSIDE / top face - all bends DOWN) scale 1:3



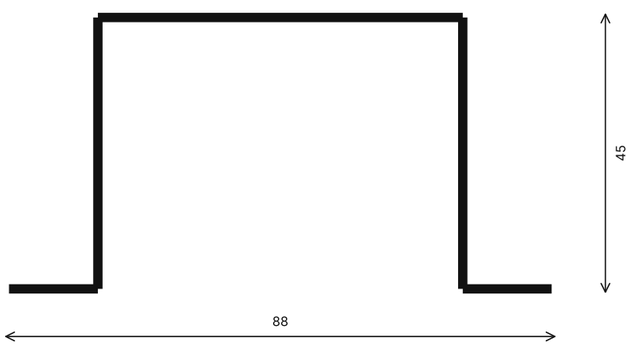
BEND TABLE

No.	Pos. from datum edge	Angle	Dir.	Inside R	Formed (outside)
1	13.70	90°	DOWN	R1.5	Foot 15
2	56.09	90°	DOWN	R1.5	Web 45
3	113.48	90°	DOWN	R1.5	Top 60
4	155.88	90°	DOWN	R1.5	Web 45
	edge 169.57				Foot 15

K-factor 0.44, BA(90°)=3.39, BD(90°)=2.61. Adjust flat to your tooling if your BD differs.

Bend sequence: 1 then 2, 3, 4 (long bends first, lip/return last). Check first-off against FORMED SECTION.

FORMED SECTION (scale 1.0:1)



Foot 15 | Web 45 | Top 60 | Web 45 | Foot 15 (outside dims)

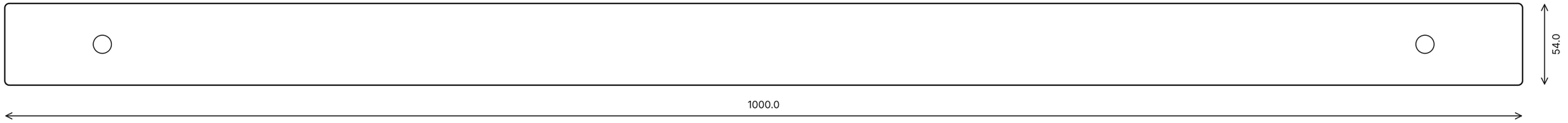
NOTES

1. Top-hat section, open side DOWN (feet rest on canopy roof skin).
2. End tabs (4 off, 37 × 1.5) pass through side-rail wall slots; weld outside and grind flush.
3. Feet notched 16 mm at ends to clear side-rail bottom lip.
4. 3 x accessory slots 50 × 10 in deck face (rated tie-down points).
5. Mount slots 30 × 11 at X = ±435.7: VERIFY against canopy roof holes before cutting.
6. Fit closing plate RRD-2005 inside hat, flush with feet, before fitting to rack (box section).
7. Material: 304 stainless 2B, 1.5 mm. Laser cut from DXF 'RRD-2004-Rev-03_Mount_Cross_Member_FLAT.dxf' (1:1, mm), layer CUT only; BEND/ROLL/NOTES layers are reference.

Unless otherwise stated: dimensions in mm. Linear ±0.5, bend angle ±1°, flat pattern per K=0.44, inside R1.5. Deburr and break sharp edges. Do not scale drawing.

COMPANY VELD ARMOUR - Hilux GD6 DC canopy roof rack		DWG NO. **RRD-2004-Rev-03**
TITLE **Mount Cross Member**	MATERIAL 304 (1.4301) 2B	THICKNESS 1.5 mm
QTY / ASSY 2	FINISH Brush / passivate after weld	MASS (each) 2.26 kg
SCALE 1:3	DATE / REV 2026-10-20 Rev-03	SHEET 6 of 9 A3

FLAT PATTERN (viewed from OUTSIDE / top face - all bends DOWN) scale 1:3



BEND TABLE

No.	Pos. from datum edge	Angle	Dir.	Inside R	Formed (outside)
-	no bends - flat part				

K-factor 0.44, BA(90°)=3.39, BD(90°)=2.61. Adjust flat to your tooling if your BD differs.

Bend sequence: 1 then - (long bends first, lip/return last). Check first-off against FORMED SECTION.

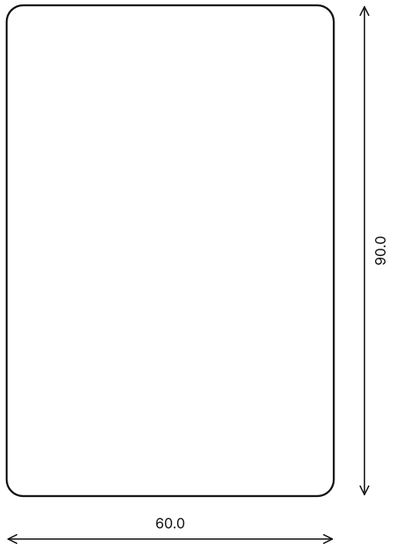
NOTES

1. Flat plate. Slides inside RRD-2004 hat, flush with feet; stitch weld 25/100 to webs from below.
2. Ø12 holes line up with mount slots; fit spacer tube between plate and roof rib (see assembly).
3. Material: 304 stainless 2B, 1.5 mm. Laser cut from DXF 'RRD-2005-Rev-03_Mount_Cross_Closing_Plate_FLAT.dxf' (1:1, mm), layer CUT only; BEND/ROLL/NOTES layers are reference.

Unless otherwise stated: dimensions in mm. Linear ±0.5, bend angle ±1°, flat pattern per K=0.44, inside R1.5. Deburr and break sharp edges. Do not scale drawing.

COMPANY VELD ARMOUR - Hilux GD6 DC canopy roof rack		DWG NO. **RRD-2005-Rev-03**
TITLE **Mount Cross Closing Plate**	MATERIAL 304 (1.4301) 2B	THICKNESS 1.5 mm
QTY / ASSY 2	FINISH Brush / passivate after weld	MASS (each) 0.64 kg
SCALE 1:3	DATE / REV 2026-10-20 Rev-03	SHEET 7 of 9 A3

FLAT PATTERN (viewed from OUTSIDE / top face - all bends DOWN) scale 1:1



BEND TABLE

No.	Pos. from datum edge	Angle	Dir.	Inside R	Formed (outside)
-	no bends - flat part				

K-factor 0.44, BA(90°)=3.39, BD(90°)=2.61. Adjust flat to your tooling if your BD differs.

Bend sequence: 1 then - (long bends first, lip/return last). Check first-off against FORMED SECTION.

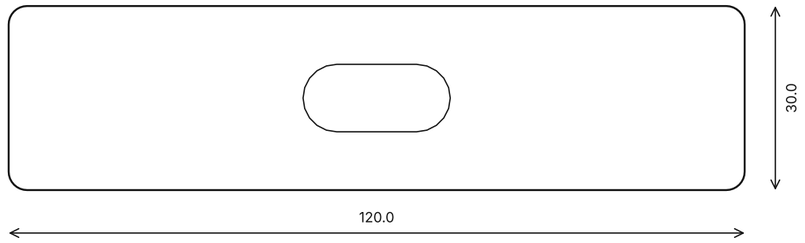
NOTES

1. Flat plate, fits inside wall behind PS/DS butt joint (front and rear centre).
2. Plug/fillet weld to wall from inside, both halves.
3. Material: 304 stainless 2B, 1.5 mm. Laser cut from DXF 'RRD-2006-Rev-03_Rail_Splice_Plate_FLAT.dxf' (1:1, mm), layer CUT only; BEND/ROLL/NOTES layers are reference.

Unless otherwise stated: dimensions in mm. Linear ±0.5, bend angle ±1°, flat pattern per K=0.44, inside R1.5. Deburr and break sharp edges. Do not scale drawing.

COMPANY VELD ARMOUR - Hilux GD6 DC canopy roof rack		DWG NO. **RRD-2006-Rev-03**
TITLE **Rail Splice Plate**	MATERIAL 304 (1.4301) 2B	THICKNESS 1.5 mm
QTY / ASSY 2	FINISH Brush / passivate after weld	MASS (each) 0.06 kg
SCALE 1:1	DATE / REV 2026-10-20 Rev-03	SHEET 8 of 9 A3

FLAT PATTERN (viewed from OUTSIDE / top face - all bends DOWN) scale 1:1



BEND TABLE

No.	Pos. from datum edge	Angle	Dir.	Inside R	Formed (outside)
-	no bends - flat part				

K-factor 0.44, BA(90°)=3.39, BD(90°)=2.61. Adjust flat to your tooling if your BD differs.

Bend sequence: 1 then - (long bends first, lip/return last). Check first-off against FORMED SECTION.

NOTES

1. Flat plate, fits inside canopy under roof rib top flat (35 wide channel).
2. Slot 24 × 11 along vehicle axis for M10 mounting bolt.
3. Material: 304 stainless 2B, 1.5 mm. Laser cut from DXF
'RRD-2007-Rev-03_Roof_Backing_Plate_FLAT.dxf' (1:1, mm), layer CUT only;
BEND/ROLL/NOTES layers are reference.

Unless otherwise stated: dimensions in mm. Linear ±0.5, bend angle ±1°, flat pattern per K=0.44, inside R1.5. Deburr and break sharp edges. Do not scale drawing.

COMPANY VELD ARMOUR - Hilux GD6 DC canopy roof rack		DWG NO. **RRD-2007-Rev-03**
TITLE **Roof Backing Plate**	MATERIAL 304 (1.4301) 2B	THICKNESS 1.5 mm
QTY / ASSY 4	FINISH Brush / passivate after weld	MASS (each) 0.04 kg
SCALE 1:1	DATE / REV 2026-10-20 Rev-03	SHEET 9 of 9 A3

Structural calculation - canopy roof rack RRD-2000-Rev-03

Veld Armour - Toyota Hilux GD6 double cab canopy (roof panel 40105). Date 2026-10-20. Prepared from the parametric design file (rack_design.py) - all numbers below are generated from the same parameters as the DXF/PDF drawing set.



1. Design basis

Rack footprint 1130 x 1350 mm, plan corners R60, wall height 100 mm, deck (top of cross members) at 45 mm above the canopy roof skin. Material 304 / 1.4301 stainless, 1.5 mm, 2B. Fabricated mass 26.6 kg.

Item	Value	Basis
Yield strength Rp0.2	205 MPa	ASTM A240 / EN 10088-2 minimum for 304 annealed
Tensile strength Rm	515 MPa	ASTM A240 minimum
Young's modulus	193 GPa	304
Rated payload	250 kg (2452 N) evenly distributed	client brief 200-250 kg; 250 used
Vertical dynamic factor	3.0 g	off-road corrugations / pot-holes (industry practice for 4x4 racks)
Longitudinal / lateral	1.5 g / 1.0 g	emergency braking / cornering
Point load	120 kg person on one cross member, mid-span	serviceability (loading the rack)
Acceptance	dynamic case: stress <= Rp0.2 with FoS >= 1.25; static deflection <= L/250	no permanent set in service

Load path: cargo -> deck cross members (7 off top-hat 60x45x1.5) -> tabs in side-wall slots -> side rails (wall 100 + boxed top rail) -> mount cross members at Y = ±470 (box section with closing plate) -> 4x M10 bolts -> canopy roof rib holes with backing plates. Cross-member feet also bear directly on the roof skin, which is conservatively ignored.

2. Section properties (thin-wall, centre-line)

Section	A mm²	Ixx mm■	Z mm³
Deck cross, top-hat 60 x 45 x 15 feet	261	78667	2997
Mount cross, hat + 54 closing plate (box)	-	118878	4796
Side rail: lip 15 / wall 100 / flange 25 / return 25	242	293418	5004

3. Deck cross member (RRD-2003)

Worst case: one cross member carries 1/4 of the payload as a UDL (cluster of jerry cans / box on one slat), simply supported between the walls, span 1127 mm. Dynamic: w = 1.63 N/mm, M = wL²/8 = 259 Nm, σ = M/Z = 86 MPa, FoS = 2.37 OK. Person load 1177 N at mid-span (static): M = 332 Nm, σ = 111 MPa, FoS = 1.85 OK. Static deflection under rated UDL 0.75 mm (L/1497), under the person 2.3 mm (L/487).

Local buckling (AISI S100 effective width): top flange b/t = 38 and web h/t = 28 are below the fully-effective limit 1.28√(E/fy) = 39, so the gross section is valid.

4. Side rail (RRD-2001 / 2002)

Each rail collects half of the payload through the cross-member tabs and spans between the two mount rows (span 940 mm, overhang 205 mm each end). Dynamic UDL w = 2.73 N/mm: M(support) = 57 Nm, M(mid-span) = 244 Nm, σ = 49 MPa, FoS = 4.21 OK. Static mid-span deflection 0.13 mm. Reaction per rail per mount row (dynamic) R = 1839 N - this is what the mount cross and tab welds must carry.

Wall slenderness h/t = 65 with the boxed top rail acting as an edge stiffener; flange b/t = 15 < 39: fully effective.

5. Mount cross member (RRD-2004 + closing plate RRD-2005)

The rail reaction R = 1839 N enters each end of the mount cross 129.3 mm outboard of the bolt (lever = half width - mount X). Cantilever moment 238 Nm plus the member's own tributary deck load (110 Nm) = 348 Nm. As a plain open hat this gives σ = 116 MPa (FoS 1.77); with the closing plate welded inside the hat (box section) σ = 73 MPa, FoS = 2.82 OK. The closing plate is therefore mandatory on the two mount crosses.

6. Tabs, slots and welds

Worst cross-member end reaction (dynamic) 920 N through two tabs 37 mm tall. Tab bearing in the wall slot: 8.3 MPa (FoS 25). Fillet welds 3 mm on the outside of the wall, both tabs, full height (74 mm total): capacity 48019 N at 0.6·Rm on the throat, FoS 52. Rail reactions at the mount rows (1839 N) go through the same tab/weld detail with FoS > 25. Weld 304 with ER308L, TIG; pickle/passivate.

7. Mounting bolts and canopy interface

4x M10 A2-70 (As = 58 mm², proof load 26.1 kN). Lateral 1.0 g gives 2452 N total shear, braking 1.5 g gives 3679 N: 920 N per bolt. Overturning from lateral load at a cargo c.g. 300 mm above the deck adds 485 N tension per bolt; a 1 g rebound/inverted case adds 613 N. Combined utilisation (tension² + shear²) = 0.005 - bolts are not critical. Clamp load spreads through the crush tube (no hat crushing) and the Ø30 washer / 120 x 30 backing plate onto the 1.5 mm roof rib: bearing 1.8 MPa. Torque 35 Nm.

The canopy roof panel 40105 and the canopy side walls carry the rack + payload (≈ 277 kg static, 830 kg dynamic) - they are outside this calculation and should be checked by Veld Armour against the canopy design, as for Rev-02.

8. Wall tie-down slots - usage limit

The 60 x 20 slots in the wall sit 34 mm above the line where the cross-member tabs brace the wall. A horizontal (inward) strap pull at a slot bends the 1.5 mm wall as a plate cantilever: with a 45° load spread (effective width 128 mm) the wall reaches yield at 289 N (≈ 30 kg). In-plane (vertical) pull is limited by shear of the wall above the slot at > 8118 N. The boxed top rail alone, as a lateral beam between the corners, yields at a 623 N mid-span point load. Therefore: wall slots are for cargo nets and light lashing only (≤ 25 kg lateral per slot); rated tie-down points are the deck accessory slots (3 per cross member, 21 total, 150 kg each - cross-member capacity governs).

9. Summary

Check	Load case	Stress MPa	FoS on Rp0.2	Result
Deck cross - bending	1/4 payload UDL x 3 g	86	2.37	OK
Deck cross - point load	120 kg person mid-span	111	1.85	OK
Side rail - bending	1/2 payload UDL x 3 g	49	4.21	OK
Mount cross (box) - bending	rail reaction + deck, 3 g	73	2.82	OK
Tab weld	cross end reaction, 3 g	-	52	OK
M10 bolts	1.5 g long. / 1 g lat. / 1 g rebound	-	util. 0.005	OK
Static deflection	250 kg, deck cross	0.75 mm	L/1497	OK
Wall slot lateral pull	strap at slot	yield at 289 N	-	usage limit

Conclusion: the Rev-03 cage-style rack in 1.5 mm 304 is adequate for a 250 kg evenly distributed payload with a 3 g vertical dynamic factor, provided (a) the two mount crosses are built as box sections with the closing plate, (b) heavy loads are strapped to the deck slots, not the wall, and (c) the canopy roof hole pitch is verified before cutting the mount crosses. Increase to 2.0 mm 304 if the rack will routinely carry more than 250 kg or a tent plus occupants (static 250 kg deck load with people on it exceeds the point-load case).

Assumptions and limits: thin-wall elastic beam theory, bend radii ignored in section properties (conservative), no fatigue assessment (stresses at 1 g are < 40 MPa, well below the 304 endurance limit); welds assumed sound and full length; canopy structure not assessed.